



National  
Qualifications  
2023

# **2023 Mathematics**

## **Higher - Paper 1**

### **Finalised Marking Instructions**

© Scottish Qualifications Authority 2023

These marking instructions have been prepared by examination teams for use by SQA appointed markers when marking external course assessments.

The information in this document may be reproduced in support of SQA qualifications only on a non-commercial basis. If it is reproduced, SQA must be clearly acknowledged as the source. If it is to be reproduced for any other purpose, written permission must be obtained from [permissions@sqa.org.uk](mailto:permissions@sqa.org.uk).



## General marking principles for Higher Mathematics

Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

For each question, the marking instructions are generally in two sections:

- *generic scheme* – this indicates why each mark is awarded
- *illustrative scheme* – this covers methods which are commonly seen throughout the marking

In general, you should use the illustrative scheme. Only use the generic scheme where a candidate has used a method not covered in the illustrative scheme.

- Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.
- If you are uncertain how to assess a specific candidate response because it is not covered by the general marking principles or the detailed marking instructions, you must seek guidance from your team leader.
- One mark is available for each •. There are no half marks.
- If a candidate's response contains an error, all working subsequent to this error must still be marked. Only award marks if the level of difficulty in their working is similar to the level of difficulty in the illustrative scheme.
- Only award full marks where the solution contains appropriate working. A correct answer with no working receives no mark, unless specifically mentioned in the marking instructions.
- Candidates may use any mathematically correct method to answer questions, except in cases where a particular method is specified or excluded.
- If an error is trivial, casual or insignificant, for example  $6 \times 6 = 12$ , candidates lose the opportunity to gain a mark, except for instances such as the second example in point (h) below.
- If a candidate makes a transcription error (question paper to script or within script), they lose the opportunity to gain the next process mark, for example

This is a transcription error and so the mark is not awarded.

This is no longer a solution of a quadratic equation, so the mark is not awarded.

$$x^2 + 5x + 7 = 9x + 4$$

$$x - 4x + 3 = 0$$

$$x = 1$$

The following example is an exception to the above

This error is not treated as a transcription error, as the candidate deals with the intended quadratic equation. The candidate has been given the benefit of the doubt and all marks awarded.

$$x^2 + 5x + 7 = 9x + 4$$

$$x - 4x + 3 = 0$$

$$(x - 3)(x - 1) = 0$$

$$x = 1 \text{ or } 3$$

(i) **Horizontal/vertical marking**

If a question results in two pairs of solutions, apply the following technique, but only if indicated in the detailed marking instructions for the question.

Example:

$$\begin{array}{cc} \bullet^5 & \bullet^6 \\ \bullet^5 & x = 2 \quad x = -4 \\ \bullet^6 & y = 5 \quad y = -7 \end{array}$$

Horizontal:  $\bullet^5 x = 2$  and  $x = -4$     Vertical:  $\bullet^5 x = 2$  and  $y = 5$   
 $\bullet^6 y = 5$  and  $y = -7$      $\bullet^6 x = -4$  and  $y = -7$

You must choose whichever method benefits the candidate, **not** a combination of both.

(j) In final answers, candidates should simplify numerical values as far as possible unless specifically mentioned in the detailed marking instruction. For example

$$\begin{array}{ll} \frac{15}{12} \text{ must be simplified to } \frac{5}{4} \text{ or } 1\frac{1}{4} & \frac{43}{1} \text{ must be simplified to } 43 \\ \frac{15}{0.3} \text{ must be simplified to } 50 & \frac{4\cancel{5}}{3} \text{ must be simplified to } \frac{4}{15} \\ \sqrt{64} \text{ must be simplified to } 8^* & \end{array}$$

\*The square root of perfect squares up to and including 144 must be known.

(k) Commonly Observed Responses (COR) are shown in the marking instructions to help mark common and/or non-routine solutions. CORs may also be used as a guide when marking similar non-routine candidate responses.

(l) Do not penalise candidates for any of the following, unless specifically mentioned in the detailed marking instructions:

- working subsequent to a correct answer
- correct working in the wrong part of a question
- legitimate variations in numerical answers/algebraic expressions, for example angles in degrees rounded to nearest degree
- omission of units
- bad form (bad form only becomes bad form if subsequent working is correct), for example  
 $(x^3 + 2x^2 + 3x + 2)(2x + 1)$  written as  
 $(x^3 + 2x^2 + 3x + 2) \times 2x + 1$   
 $= 2x^4 + 5x^3 + 8x^2 + 7x + 2$  gains full credit
- repeated error within a question, but not between questions or papers

(m) In any 'Show that...' question, where candidates have to arrive at a required result, the last mark is not awarded as a follow-through from a previous error, unless specified in the detailed marking instructions.

- (n) You must check all working carefully, even where a fundamental misunderstanding is apparent early in a candidate's response. You may still be able to award marks later in the question so you must refer continually to the marking instructions. The appearance of the correct answer does not necessarily indicate that you can award all the available marks to a candidate.
- (o) You should mark legible scored-out working that has not been replaced. However, if the scored-out working has been replaced, you must only mark the replacement working.
- (p) If candidates make multiple attempts using the same strategy and do not identify their final answer, mark all attempts and award the lowest mark. If candidates try different valid strategies, apply the above rule to attempts within each strategy and then award the highest mark.

For example:

|                                                                    |                                                                    |
|--------------------------------------------------------------------|--------------------------------------------------------------------|
| Strategy 1 attempt 1 is worth 3 marks.                             | Strategy 2 attempt 1 is worth 1 mark.                              |
| Strategy 1 attempt 2 is worth 4 marks.                             | Strategy 2 attempt 2 is worth 5 marks.                             |
| From the attempts using strategy 1, the resultant mark would be 3. | From the attempts using strategy 2, the resultant mark would be 1. |

In this case, award 3 marks.

# Marking instructions for each question

| Question                                                                                                                                                                                                                                                      |  |  | Generic scheme                                                                                                                                                                                           | Illustrative scheme                                                                                                                                                                                                                                                                                | Max mark |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1.                                                                                                                                                                                                                                                            |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> express second term in differentiable form</li> <li>•<sup>2</sup> differentiate one term</li> <li>•<sup>3</sup> complete differentiation</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> ... - 10x<sup>-4</sup> stated or implied by •<sup>3</sup></li> <li>•<sup>2</sup> <math>\frac{5}{3}x^{\frac{2}{3}} \dots</math> or ... + 40x<sup>-5</sup></li> <li>•<sup>3</sup> <math>\frac{5}{3}x^{\frac{2}{3}} + 40x^{-5}</math></li> </ul> | 3        |
| Notes:                                                                                                                                                                                                                                                        |  |  |                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                    |          |
| 1. Where candidates “differentiate over two lines” see Candidates A and B.<br>2. • <sup>3</sup> is only available for differentiating a term with a negative index.<br>3. Where candidates attempt to integrate throughout, only • <sup>1</sup> is available. |  |  |                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                    |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                  |  |  |                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                    |          |
| Candidate A - differentiating over two lines                                                                                                                                                                                                                  |  |  | Candidate B - differentiating over two lines                                                                                                                                                             |                                                                                                                                                                                                                                                                                                    |          |
| $y = x^{\frac{5}{3}} - \frac{10}{x^4}$<br>$y = \frac{5}{3}x^{\frac{2}{3}} - 10x^{-4}$ • <sup>1</sup> ✓<br>$y = \frac{5}{3}x^{\frac{2}{3}} + 40x^{-5}$ • <sup>2</sup> ✓ • <sup>3</sup> ✗                                                                       |  |  | $y = x^{\frac{5}{3}} - \frac{10}{x^4}$<br>$y = \frac{5}{3}x^{\frac{2}{3}} - 10x^{-4}$ • <sup>1</sup> ✓<br>$y = \frac{5}{3}x^{\frac{2}{3}} + 40x^{-3}$ • <sup>2</sup> ✓ • <sup>3</sup> ✗                  |                                                                                                                                                                                                                                                                                                    |          |
| Candidate C<br>∴<br>$\frac{5}{3}x^{\frac{2}{3}} + 40x^{-5} + c$ • <sup>3</sup> ✗                                                                                                                                                                              |  |  |                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                    |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                |  |  | Generic scheme                                                                                                                                                                                                                                              | Illustrative scheme                                                                                                                                                                                                                                             | Max mark |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2.                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> find midpoint of PQ</li> <li>•<sup>2</sup> calculate gradient of PQ</li> <li>•<sup>3</sup> state perpendicular gradient</li> <li>•<sup>4</sup> determine equation of perpendicular bisector</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> (4,3)</li> <li>•<sup>2</sup> <math>-\frac{1}{2}</math> or <math>-\frac{6}{12}</math></li> <li>•<sup>3</sup> 2 stated or implied by •<sup>4</sup></li> <li>•<sup>4</sup> <math>y = 2x - 5</math></li> </ul> | 4        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                 |          |
| 1. • <sup>4</sup> is only available as a consequence of using a perpendicular gradient <b>and</b> a midpoint.<br>2. The gradient of the perpendicular bisector must appear in fully simplified form at • <sup>3</sup> or • <sup>4</sup> stage for • <sup>4</sup> to be awarded.<br>3. At • <sup>4</sup> , accept $2x - y = 5$ , $y - 2x = -5$ or any other rearrangement of the equation where the constant terms have been simplified. |  |  |                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                 |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                 |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |  |                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                 |          |

| Question                                                                                                                                                                                                                                                   |  |  | Generic scheme                                                                                                                                                                                                                                                  | Illustrative scheme                                                                                                                                                                                    | Max mark |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3.                                                                                                                                                                                                                                                         |  |  | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> apply <math>\log_5 x - \log_5 y = \log_5 \frac{x}{y}</math></li> <li>•<sup>2</sup> write in exponential form</li> <li>•<sup>3</sup> process for <math>x</math></li> </ul>                  | <b>Method 1</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\log_5 \frac{x}{3} \dots</math></li> <li>•<sup>2</sup> <math>\frac{x}{3} = 5^2</math></li> <li>•<sup>3</sup> 75</li> </ul>  | 3        |
|                                                                                                                                                                                                                                                            |  |  | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> apply <math>\log_5 x - \log_5 y = \log_5 \frac{x}{y}</math></li> <li>•<sup>2</sup> apply <math>m \log_5 x = \log_5 x^m</math></li> <li>•<sup>3</sup> process for <math>x</math></li> </ul> | <b>Method 2</b> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\log_5 \frac{x}{3} \dots</math></li> <li>•<sup>2</sup> <math>\dots = \log_5 5^2</math></li> <li>•<sup>3</sup> 75</li> </ul> | 3        |
| <b>Notes:</b>                                                                                                                                                                                                                                              |  |  |                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                        |          |
| 1. Each line of working must be equivalent to the line above within a valid strategy, however see Candidates A and B for exceptions.<br>2. Where candidates do not use exponentials at • <sup>2</sup> , • <sup>3</sup> is not available - see Candidate C. |  |  |                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                        |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                        |  |  |                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                        |          |
| <b>Candidate A - incorrect exponential</b>                                                                                                                                                                                                                 |  |  | <b>Candidate B</b>                                                                                                                                                                                                                                              |                                                                                                                                                                                                        |          |
| $\log_5 \frac{x}{3} = 2$                                                                                                                                                                                                                                   |  |  | $\log_5 3x = 2$                                                                                                                                                                                                                                                 |                                                                                                                                                                                                        |          |
| • <sup>1</sup> ✓                                                                                                                                                                                                                                           |  |  | • <sup>1</sup> ✗                                                                                                                                                                                                                                                |                                                                                                                                                                                                        |          |
| $\frac{x}{3} = 2^5$                                                                                                                                                                                                                                        |  |  | $3x = 5^2$                                                                                                                                                                                                                                                      |                                                                                                                                                                                                        |          |
| • <sup>2</sup> ✗                                                                                                                                                                                                                                           |  |  | • <sup>2</sup> ✓ <sub>1</sub>                                                                                                                                                                                                                                   |                                                                                                                                                                                                        |          |
| $x = 96$                                                                                                                                                                                                                                                   |  |  | $x = \frac{25}{3}$                                                                                                                                                                                                                                              |                                                                                                                                                                                                        |          |
| • <sup>3</sup> ✓ <sub>1</sub>                                                                                                                                                                                                                              |  |  | • <sup>3</sup> ✓ <sub>1</sub>                                                                                                                                                                                                                                   |                                                                                                                                                                                                        |          |
| <b>Candidate C - no use of exponentials</b>                                                                                                                                                                                                                |  |  |                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                        |          |
| $\log_5 \frac{x}{3} = 2$                                                                                                                                                                                                                                   |  |  |                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                        |          |
| • <sup>1</sup> ✓                                                                                                                                                                                                                                           |  |  |                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                        |          |
| $\frac{x}{3} = 10$                                                                                                                                                                                                                                         |  |  |                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                        |          |
| • <sup>2</sup> ✗                                                                                                                                                                                                                                           |  |  |                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                        |          |
| $x = 30$                                                                                                                                                                                                                                                   |  |  |                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                        |          |
| • <sup>3</sup> ✗                                                                                                                                                                                                                                           |  |  |                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                        |          |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |  | Generic scheme                                                                                                                                                          | Illustrative scheme                                                                                                                                                                                                              | Max mark |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | (a) |  | • <sup>1</sup> find $\cos p$                                                                                                                                            | • <sup>1</sup> $\frac{3}{5}$                                                                                                                                                                                                     | 1        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |  | • <sup>2</sup> find $\cos q$                                                                                                                                            | • <sup>2</sup> $\frac{3}{\sqrt{45}} \left( = \frac{1}{\sqrt{5}} \right)$                                                                                                                                                         | 1        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |  |                                                                                                                                                                         |                                                                                                                                                                                                                                  |          |
| 1. Accept $\frac{3}{3\sqrt{5}}$ for • <sup>2</sup> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |  |                                                                                                                                                                         |                                                                                                                                                                                                                                  |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |  |                                                                                                                                                                         |                                                                                                                                                                                                                                  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | (b) |  | • <sup>3</sup> select appropriate formula and express in terms of $p$ and $q$<br>• <sup>4</sup> substitute into addition formula<br>• <sup>5</sup> evaluate $\cos(p+q)$ | • <sup>3</sup> $\cos p \cos q - \sin p \sin q$<br>• <sup>4</sup> $\frac{3}{5} \times \frac{3}{\sqrt{45}} - \frac{4}{5} \times \frac{6}{\sqrt{45}}$<br>• <sup>5</sup> $-\frac{3}{\sqrt{45}} \left( = -\frac{1}{\sqrt{5}} \right)$ | 3        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |  |                                                                                                                                                                         |                                                                                                                                                                                                                                  |          |
| 2. Award • <sup>3</sup> for candidates who write $\cos\left(\frac{3}{5}\right) \times \cos\left(\frac{3}{\sqrt{45}}\right) - \sin\left(\frac{4}{5}\right) \times \sin\left(\frac{6}{\sqrt{45}}\right)$ . • <sup>4</sup> and • <sup>5</sup> are unavailable.<br>3. For any attempt to use $\cos(p+q) = \cos p \pm \cos q$ , • <sup>4</sup> and • <sup>5</sup> are unavailable.<br>4. • <sup>5</sup> is only available if either the surd part or the non-surd part of the fraction is simplified as far as possible. Accept $-\frac{3}{\sqrt{45}}$ , $-\frac{\sqrt{45}}{15}$ , $-\frac{15}{15\sqrt{5}}$ or answers obtained on follow through which do not require simplification. Do not accept $-\frac{15}{5\sqrt{45}}$ .<br>5. • <sup>5</sup> is only available for an answer expressed as a single fraction. |     |  |                                                                                                                                                                         |                                                                                                                                                                                                                                  |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |  |                                                                                                                                                                         |                                                                                                                                                                                                                                  |          |



| Question                                                                                                                                                                                                                                                      |  |  | Generic scheme                                                                                                                                                                                                       | Illustrative scheme                                                                                                                                                                                                   | Max mark                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 5.                                                                                                                                                                                                                                                            |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> use the discriminant</li> <li>•<sup>2</sup> apply condition and express in standard quadratic form</li> <li>•<sup>3</sup> process for <math>p</math></li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>(3p-2)^2 - 4 \times 2 \times p</math></li> <li>•<sup>2</sup> <math>9p^2 - 20p + 4 = 0</math></li> <li>•<sup>3</sup> <math>\frac{2}{9}, 2</math></li> </ul> | 3                             |
| <b>Notes:</b>                                                                                                                                                                                                                                                 |  |  |                                                                                                                                                                                                                      |                                                                                                                                                                                                                       |                               |
| 1. Where candidates states an incorrect condition, • <sup>2</sup> is not available. However, • <sup>3</sup> is available for finding the roots of the quadratic - see Candidate B.<br>2. Where $x$ appears in any expression, no further marks are available. |  |  |                                                                                                                                                                                                                      |                                                                                                                                                                                                                       |                               |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                           |  |  |                                                                                                                                                                                                                      |                                                                                                                                                                                                                       |                               |
| <b>Candidate A</b>                                                                                                                                                                                                                                            |  |  | <b>Candidate B</b>                                                                                                                                                                                                   |                                                                                                                                                                                                                       |                               |
| (For equal roots) $b^2 - 4ac = 0$                                                                                                                                                                                                                             |  |  | (For equal roots) $b^2 - 4ac > 0$                                                                                                                                                                                    |                                                                                                                                                                                                                       |                               |
| $(3p-2)^2 - 4 \times 2 \times p$                                                                                                                                                                                                                              |  |  | $(3p-2)^2 - 4 \times 2 \times p$                                                                                                                                                                                     | • <sup>2</sup> ✗                                                                                                                                                                                                      | • <sup>1</sup> ✓              |
| $9p^2 - 20p + 4$                                                                                                                                                                                                                                              |  |  | $9p^2 - 20p + 4 = 0$                                                                                                                                                                                                 |                                                                                                                                                                                                                       |                               |
| $p = \frac{2}{9}, 2$                                                                                                                                                                                                                                          |  |  | $p = \frac{2}{9}, 2$                                                                                                                                                                                                 | • <sup>3</sup> ✓                                                                                                                                                                                                      | • <sup>3</sup> ✓ <sub>1</sub> |
|                                                                                                                                                                                                                                                               |  |  | ⋮                                                                                                                                                                                                                    |                                                                                                                                                                                                                       |                               |

| Question |  |  | Generic scheme                                                                                                                                                                                                                           | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                       | Max mark |
|----------|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 6.       |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> express second term in integrable form</li> <li>•<sup>2</sup> integrate one term</li> <li>•<sup>3</sup> integrate other term</li> <li>•<sup>4</sup> complete integration</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> ... - 6x<sup><math>\frac{1}{2}</math></sup></li> <li>•<sup>2</sup> <math>\frac{2}{6}x^6</math> ... or ... - <math>\frac{6x^{\frac{3}{2}}}{\frac{3}{2}}</math></li> <li>•<sup>3</sup> ... - <math>\frac{6x^{\frac{3}{2}}}{\frac{3}{2}}</math> or <math>\frac{2}{6}x^6</math> ...</li> <li>•<sup>4</sup> <math>\frac{1}{3}x^6 - 4x^{\frac{3}{2}} + c</math></li> </ul> | 4        |

#### Notes:

- The mark for integrating the final term is only available if candidates integrate a term with a fractional index.
- Do not penalise the appearance of an integral sign and/or  $dx$  throughout.
- Do not penalise the omission of '+c' at •<sup>2</sup> or •<sup>3</sup>.
- All coefficients must be simplified at •<sup>4</sup> stage for •<sup>4</sup> to be awarded.
- Accept  $\frac{x^6 - 12x^{\frac{3}{2}}}{3} + c$  for •<sup>4</sup> but do not accept  $\frac{2x^6 - 24x^{\frac{3}{2}}}{6} + c$ .
- <sup>2</sup>, •<sup>3</sup> and •<sup>4</sup> are not available within an invalid strategy.

#### Commonly Observed Responses:

|                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Candidate A</b><br>$\int (2x^5 - 6x^{\frac{1}{2}}) dx$ • <sup>1</sup> ✓<br>$= \frac{2x^6}{6} - \frac{6x^{\frac{3}{2}}}{\frac{3}{2}} + c$ • <sup>2</sup> ✓ • <sup>3</sup> ✓<br>$= \frac{2x^6}{6} - 4x^{\frac{3}{2}} + c$ • <sup>4</sup> ✗<br>$= \frac{1}{3}x^6 - 4\sqrt{x^3} + c$ • <sup>4</sup> ✗<br>• <sup>4</sup> cannot be awarded over two lines of working | <b>Candidate B - integrating over two lines</b><br>$\frac{2x^6}{6} - 6x^{\frac{1}{2}}$ • <sup>1</sup> ✓<br>$= \frac{2x^6}{6} - \frac{6x^{\frac{3}{2}}}{\frac{3}{2}} + c$ • <sup>2</sup> ✓ • <sup>3</sup> ✗<br>$= \frac{1}{3}x^6 - 4x^{\frac{3}{2}} + c$ • <sup>4</sup> ✓ <sub>1</sub> |
| <b>Candidate C - insufficient evidence</b><br>$\int 2x^5 - 6x^{\frac{1}{2}} dx$ • <sup>1</sup> ✓<br>$\frac{1}{3}x^6 - 9x^{\frac{3}{2}} + c$ • <sup>2</sup> ✓ • <sup>3</sup> ✗ • <sup>4</sup> ✗                                                                                                                                                                     | <b>Candidate D</b><br>$\vdots$ • <sup>1</sup> ✓<br>$= \frac{1}{3}x^6 - 4x^{\frac{3}{2}}$ • <sup>2</sup> ✓ • <sup>3</sup> ✓<br>$= \frac{1}{3}x^6 - 4\sqrt{x^3} + c$ • <sup>4</sup> ✓                                                                                                   |

| Question                                                                                                                                                                                  |     |  | Generic scheme                                                                                                       | Illustrative scheme                                                                                                                     | Max mark |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7.                                                                                                                                                                                        | (a) |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> use laws of logs</li> <li>•<sup>2</sup> evaluate log</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\log_2 \frac{5}{40}</math></li> <li>•<sup>2</sup> <math>-3</math></li> </ul> | 2        |
| <b>Notes:</b>                                                                                                                                                                             |     |  |                                                                                                                      |                                                                                                                                         |          |
| 1. Do not penalise the omission of the base of the logarithm at • <sup>1</sup> .<br>2. Correct answer with no working, award 0/2.                                                         |     |  |                                                                                                                      |                                                                                                                                         |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                       |     |  |                                                                                                                      |                                                                                                                                         |          |
| <b>Candidate A - introducing a variable</b><br>$\log_2 \left( 5 \times \frac{1}{40} \right)$ • <sup>1</sup> ✓<br>$\log_2 \frac{1}{8}$<br>$2^x = \frac{1}{8}$<br>$x = -3$ • <sup>2</sup> ✓ |     |  |                                                                                                                      |                                                                                                                                         |          |
|                                                                                                                                                                                           | (b) |  | • <sup>3</sup> state range                                                                                           | • <sup>3</sup> $0 < a < 1$                                                                                                              | 1        |
| <b>Notes:</b>                                                                                                                                                                             |     |  |                                                                                                                      |                                                                                                                                         |          |
| 3. At • <sup>3</sup> accept “ $a > 0$ and $a < 1$ ” or “ $a > 0, a < 1$ ”.                                                                                                                |     |  |                                                                                                                      |                                                                                                                                         |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                       |     |  |                                                                                                                      |                                                                                                                                         |          |
|                                                                                                                                                                                           |     |  |                                                                                                                      |                                                                                                                                         |          |

| Question |           |            | Generic scheme                                                                                                                                                                                                                                                                                                                                            | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Max mark  |         |      |         |     |         |         |     |     |     |     |     |       |           |            |             |            |           |   |
|----------|-----------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------|------|---------|-----|---------|---------|-----|-----|-----|-----|-----|-------|-----------|------------|-------------|------------|-----------|---|
| 8.       |           |            | <ul style="list-style-type: none"><li>•<sup>1</sup> start to differentiate</li><li>•<sup>2</sup> complete differentiation and equate to 0</li><li>•<sup>3</sup> solve for <math>x</math></li><li>•<sup>4</sup> process for <math>y</math></li><li>•<sup>5</sup> construct nature table(s)</li><li>•<sup>6</sup> interpret and state conclusions</li></ul> | <ul style="list-style-type: none"><li>•<sup>1</sup> <math>3x^2 \dots</math> or <math>\dots + 6x \dots</math> or <math>\dots - 9</math></li><li>•<sup>2</sup> <math>3x^2 + 6x - 9 = 0</math></li><li>•<sup>3</sup> <math>-3</math> and <math>1</math></li><li>•<sup>4</sup> <math>32</math> and <math>0</math></li><li>•<sup>5</sup><table><tr><td><math>x</math></td><td><math>\dots</math></td><td><math>-3</math></td><td><math>\dots</math></td><td><math>1</math></td><td><math>\dots</math></td></tr><tr><td><math>f'(x)</math></td><td><math>+</math></td><td><math>0</math></td><td><math>-</math></td><td><math>0</math></td><td><math>+</math></td></tr><tr><td>shape</td><td><math>\diagup</math></td><td><math>\text{—}</math></td><td><math>\diagdown</math></td><td><math>\text{—}</math></td><td><math>\diagup</math></td></tr></table></li><li>•<sup>6</sup> max at <math>(-3, 32)</math>; min at <math>(1, 0)</math></li></ul> | $x$       | $\dots$ | $-3$ | $\dots$ | $1$ | $\dots$ | $f'(x)$ | $+$ | $0$ | $-$ | $0$ | $+$ | shape | $\diagup$ | $\text{—}$ | $\diagdown$ | $\text{—}$ | $\diagup$ | 6 |
| $x$      | $\dots$   | $-3$       | $\dots$                                                                                                                                                                                                                                                                                                                                                   | $1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\dots$   |         |      |         |     |         |         |     |     |     |     |     |       |           |            |             |            |           |   |
| $f'(x)$  | $+$       | $0$        | $-$                                                                                                                                                                                                                                                                                                                                                       | $0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $+$       |         |      |         |     |         |         |     |     |     |     |     |       |           |            |             |            |           |   |
| shape    | $\diagup$ | $\text{—}$ | $\diagdown$                                                                                                                                                                                                                                                                                                                                               | $\text{—}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\diagup$ |         |      |         |     |         |         |     |     |     |     |     |       |           |            |             |            |           |   |

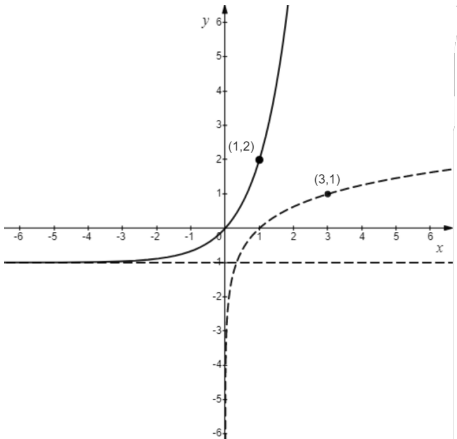
#### Notes:

- For a numerical approach award 0/6.
- <sup>2</sup> is only available if ' $= 0$ ' appears at the •<sup>2</sup> stage or in working leading to •<sup>3</sup>, however see Candidates A and B.
- Candidates who equate their derivative to 0, may use division by 3 as a strategy - see candidates B, C and D.
- <sup>3</sup> is available to candidates who factorise **their** derivative from •<sup>2</sup> as long as it is of equivalent difficulty.
- <sup>3</sup> and •<sup>4</sup> may be awarded vertically.
- <sup>5</sup> is not available where any errors are made in calculating values of  $f'(x)$ .
- <sup>5</sup> and •<sup>6</sup> may be awarded vertically.
- <sup>6</sup> is still available in cases where a candidates table of signs does not lead legitimately to a maximum/minimum shape.
- Candidates may use the second derivative - see Candidates E and F.
- Accept "max when  $x = -3$ " and "min when  $x = 1$ " for •<sup>6</sup>.

#### Commonly Observed Responses:

|                                                                                                                                                                                |                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Candidate A</b><br>Stationary points when $f'(x) = 0$<br>$f'(x) = 3x^2 + 6x - 9$ • <sup>1</sup> ✓ • <sup>2</sup> ✓<br>$f'(x) = 3(x+3)(x-1)$<br>$x = -3, 1$ • <sup>3</sup> ✓ | <b>Candidate B</b><br>Stationary points when $f'(x) = 0$<br>$f'(x) = 3x^2 + 6x - 9$ • <sup>1</sup> ✓ • <sup>2</sup> ✓<br>$\vdots$<br>$f'(x) = (x+3)(x-1)$<br>$x = -3, 1$ • <sup>3</sup> ✓                                           |
| <b>Candidate C - division by 3</b><br>$3x^2 + 6x - 9 = 0$ • <sup>1</sup> ✓ • <sup>2</sup> ✓<br>$x^2 + 2x - 3 = 0$<br>$x = -3, 1$ • <sup>3</sup> ✓                              | <b>Candidate D - derivative never equated to 0</b><br>$3x^2 + 6x - 9$ • <sup>1</sup> ✓ • <sup>2</sup> ^<br>$x^2 + 2x - 3 = 0$<br>$x = -3, 1$ • <sup>3</sup> <span style="border: 1px solid red; padding: 2px;">✓<sub>1</sub></span> |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Generic scheme                                                                        | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                          | Max mark                                                                              |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------|---------------|---------|---|----------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------|---------------|------|---------------|---------|---|----------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----|---------------|-----|---------------|---------|---|---|---|----------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 8.(continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                       |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                       |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| Candidate E - second derivative<br>$f''(x) = 6x + 6$<br><div><div><div>•<sup>5</sup> ✓</div><math>f''(-3) &lt; 0</math><br/>so max at <math>(-3, 32)</math></div><div><div>•<sup>6</sup> ✓</div><math>f''(1) &gt; 0</math><br/>so min at <math>(1, 0)</math></div><div><div>•<sup>5</sup> ✓</div><div>•<sup>6</sup> ✓</div></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                       | Candidate F - second derivative<br>$f''(x) = 6x + 6$<br><div><div><div>•<sup>5</sup> ✓</div><math>f''(-3) = -12</math>,<br/><math>-12 &lt; 0</math><br/>so max at <math>(-3, 32)</math></div><div><div>•<sup>6</sup> ✓</div><math>f''(1) = 12</math><br/><math>12 &gt; 0</math><br/>so min at <math>(1, 0)</math></div><div><div>•<sup>5</sup> ✓</div><div>•<sup>6</sup> ✓</div></div></div> |                                                                                       |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| For the table of signs for a derivative, accept:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                       |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| <table><tr><td><math>x</math></td><td><math>-3^-</math></td><td><math>-3</math></td><td><math>-3^+</math></td></tr><tr><td><math>f'(x)</math></td><td>+</td><td>0</td><td>-</td></tr><tr><td>Slope or shape</td><td></td><td></td><td></td></tr></table> <p>AND</p> <table><tr><td><math>x</math></td><td><math>1^-</math></td><td><math>1</math></td><td><math>1^+</math></td></tr><tr><td><math>f'(x)</math></td><td>-</td><td>0</td><td>+</td></tr><tr><td>Slope or shape</td><td></td><td></td><td></td></tr></table>                                                                                                                                              |                                                                                       | $x$                                                                                                                                                                                                                                                                                                                                                                                          | $-3^-$                                                                                | $-3$                                                                                  | $-3^+$                                                                                | $f'(x)$ | +             | 0       | - | Slope or shape |  |  |    | $x$                                                                                 | $1^-$                                                                               | $1$                                                                                 | $1^+$                                                                               | $f'(x)$                                                                             | -                                                                                   | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | + | Slope or shape |  |  |    | <table><tr><td><math>x</math></td><td><math>\rightarrow</math></td><td><math>-3</math></td><td><math>\rightarrow</math></td></tr><tr><td><math>f'(x)</math></td><td>+</td><td>0</td><td>-</td></tr><tr><td>Slope or shape</td><td></td><td></td><td></td></tr></table> <p>Arrows are taken to mean 'in the neighbourhood of'</p> <p>AND</p> <table><tr><td><math>x</math></td><td><math>\rightarrow</math></td><td><math>1</math></td><td><math>\rightarrow</math></td></tr><tr><td><math>f'(x)</math></td><td>-</td><td>0</td><td>+</td></tr><tr><td>Slope or shape</td><td></td><td></td><td></td></tr></table> <p>Arrows are taken to mean 'in the neighbourhood of'</p> |                                                                                       | $x$     | $\rightarrow$ | $-3$ | $\rightarrow$ | $f'(x)$ | + | 0              | -                                                                                   | Slope or shape                                                                       |      |      |      | $x$ | $\rightarrow$ | $1$ | $\rightarrow$ | $f'(x)$ | - | 0 | + | Slope or shape |  |  |  |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $-3^-$                                                                                | $-3$                                                                                                                                                                                                                                                                                                                                                                                         | $-3^+$                                                                                |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| $f'(x)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | +                                                                                     | 0                                                                                                                                                                                                                                                                                                                                                                                            | -                                                                                     |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| Slope or shape                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |      |                                                                                                                                                                                                                                                                                                             |      |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $1^-$                                                                                 | $1$                                                                                                                                                                                                                                                                                                                                                                                          | $1^+$                                                                                 |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| $f'(x)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | -                                                                                     | 0                                                                                                                                                                                                                                                                                                                                                                                            | +                                                                                     |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| Slope or shape                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |                                                                                                                                                                                                                                                                                                           |    |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $\rightarrow$                                                                         | $-3$                                                                                                                                                                                                                                                                                                                                                                                         | $\rightarrow$                                                                         |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| $f'(x)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | +                                                                                     | 0                                                                                                                                                                                                                                                                                                                                                                                            | -                                                                                     |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| Slope or shape                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |      |                                                                                                                                                                                                                                                                                                             |      |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $\rightarrow$                                                                         | $1$                                                                                                                                                                                                                                                                                                                                                                                          | $\rightarrow$                                                                         |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| $f'(x)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | -                                                                                     | 0                                                                                                                                                                                                                                                                                                                                                                                            | +                                                                                     |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| Slope or shape                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |                                                                                                                                                                                                                                                                                                           |    |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| <table><tr><td><math>x</math></td><td><math>a</math></td><td><math>-3</math></td><td><math>b</math></td></tr><tr><td><math>f'(x)</math></td><td>+</td><td>0</td><td>-</td></tr><tr><td>Slope or shape</td><td></td><td></td><td></td></tr></table> <p>Where <math>a &lt; -3</math> and <math>-3 &lt; b &lt; 1</math></p> <p>AND</p> <table><tr><td><math>x</math></td><td><math>c</math></td><td><math>1</math></td><td><math>d</math></td></tr><tr><td><math>f'(x)</math></td><td>-</td><td>0</td><td>+</td></tr><tr><td>Slope or shape</td><td></td><td></td><td></td></tr></table> <p>Where <math>-3 &lt; c &lt; 1</math> and <math>d &gt; 1</math></p> |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                       | $x$                                                                                   | $a$                                                                                   | $-3$    | $b$           | $f'(x)$ | + | 0              | -                                                                                 | Slope or shape                                                                    |  |  |  | $x$                                                                                 | $c$                                                                                 | $1$                                                                                 | $d$                                                                                 | $f'(x)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | - | 0              | +                                                                                   | Slope or shape                                                                      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $a$                                                                                   | $-3$                                                                                                                                                                                                                                                                                                                                                                                         | $b$                                                                                   |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| $f'(x)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | +                                                                                     | 0                                                                                                                                                                                                                                                                                                                                                                                            | -                                                                                     |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| Slope or shape                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |                                                                                                                                                                                                                                                                                                           |    |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $c$                                                                                   | $1$                                                                                                                                                                                                                                                                                                                                                                                          | $d$                                                                                   |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| $f'(x)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | -                                                                                     | 0                                                                                                                                                                                                                                                                                                                                                                                            | +                                                                                     |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| Slope or shape                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                                                                                                                                                                                                                                                                                                         |  |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| For the table of signs for a derivative, accept:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                       |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| <table><tr><td><math>x</math></td><td><math>\rightarrow</math></td><td><math>-3</math></td><td><math>\rightarrow</math></td><td><math>1</math></td><td><math>\rightarrow</math></td></tr><tr><td><math>f'(x)</math></td><td>+</td><td>0</td><td>-</td><td>0</td><td>+</td></tr><tr><td>Slope or shape</td><td></td><td></td><td></td><td></td><td></td></tr></table> <p>Since the function is continuous <math>-3 \rightarrow 1</math> is acceptable</p>                                                                                                                                                                                                                                                                                            |                                                                                       | $x$                                                                                                                                                                                                                                                                                                                                                                                          | $\rightarrow$                                                                         | $-3$                                                                                  | $\rightarrow$                                                                         | $1$     | $\rightarrow$ | $f'(x)$ | + | 0              | -                                                                                 | 0                                                                                 | +                                                                                   | Slope or shape                                                                      |  |  |  |  |  | <table><tr><td><math>x</math></td><td><math>a</math></td><td><math>-3</math></td><td><math>b</math></td><td><math>1</math></td><td><math>c</math></td></tr><tr><td><math>f'(x)</math></td><td>+</td><td>0</td><td>-</td><td>0</td><td>+</td></tr><tr><td>Slope or shape</td><td></td><td></td><td></td><td></td><td></td></tr></table> <p>Since the function is continuous <math>-3 &lt; b &lt; 1</math> is acceptable</p> |   | $x$            | $a$                                                                                 | $-3$                                                                                | $b$                                                                                   | $1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $c$                                                                                   | $f'(x)$ | +             | 0    | -             | 0       | + | Slope or shape |  |  |  |  |  |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $\rightarrow$                                                                         | $-3$                                                                                                                                                                                                                                                                                                                                                                                         | $\rightarrow$                                                                         | $1$                                                                                   | $\rightarrow$                                                                         |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| $f'(x)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | +                                                                                     | 0                                                                                                                                                                                                                                                                                                                                                                                            | -                                                                                     | 0                                                                                     | +                                                                                     |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| Slope or shape                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |                                                                                                                                                                                                                                                                                                           |    |    |    |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $a$                                                                                   | $-3$                                                                                                                                                                                                                                                                                                                                                                                         | $b$                                                                                   | $1$                                                                                   | $c$                                                                                   |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| $f'(x)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | +                                                                                     | 0                                                                                                                                                                                                                                                                                                                                                                                            | -                                                                                     | 0                                                                                     | +                                                                                     |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| Slope or shape                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |                                                                                                                                                                                                                                                                                                          |  |  |  |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |
| <ul style="list-style-type: none"><li>For this question do not penalise the omission of 'x' or the word 'shape'/'slope'.</li><li>Stating values of <math>f'(x)</math> is an acceptable alternative to writing '+' or '-' signs.</li><li>Acceptable variations of <math>f'(x)</math> are: <math>f'</math>, <math>\frac{df}{dx}</math>, <math>\frac{dy}{dx}</math>, <math>3x^2 + 6x - 9</math> and <math>3(x + 3)(x - 1)</math> but NOT <math>x^2 + 2x - 3</math> or <math>(x + 3)(x - 1)</math>.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                       |                                                                                       |                                                                                       |         |               |         |   |                |                                                                                   |                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |                |                                                                                     |                                                                                     |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |         |               |      |               |         |   |                |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |     |               |     |               |         |   |   |   |                |                                                                                     |                                                                                     |                                                                                     |

| Question |  |  | Generic scheme                                                                                                                                                                                                                                                                         | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                               | Max mark |
|----------|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 9.       |  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> graph reflected in <math>y = x</math></li> <li>•<sup>2</sup> vertical translation of “-1” unit following a reflection in <math>y = x</math> identifiable from graph</li> <li>•<sup>3</sup> sketch of required function</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> a concave up curve above the <math>x</math>-axis for <math>x &gt; 0</math></li> <li>•<sup>2</sup> curve passing through (0,0) and (1,2)</li> <li>•<sup>3</sup> curve approaches the line <math>y = -1</math> from above as <math>x \rightarrow -\infty</math></li> </ul>  | 3        |

#### Notes:

1. For •<sup>1</sup> accept any graph of a function which is concave up within the first quadrant.
2. •<sup>1</sup> is only available where the candidate has attempted to reflect the given curve in the line  $y = x$ .
3. •<sup>3</sup> is only available where the curve passes through (0,0) and (1,2).
4. The line  $y = -1$  does not need to be shown.
5. For a rotation, award 0/3 - for example see Candidate D.

| Question                                                           | Generic scheme | Illustrative scheme                                    | Max mark |
|--------------------------------------------------------------------|----------------|--------------------------------------------------------|----------|
| 9.(continued)                                                      |                |                                                        |          |
| Commonly Observed Responses:                                       |                |                                                        |          |
| Candidate A - reflection only                                      |                | Candidate B - translation only                         |          |
| <p>•<sup>1</sup> ✓ •<sup>2</sup> ✗ •<sup>3</sup> ✗</p>             |                | <p>•<sup>1</sup> ✗ •<sup>2</sup> ✗ •<sup>3</sup> ✗</p> |          |
| Candidate C - incorrect order of transformations                   |                | Candidate D - rotation                                 |          |
| <p>•<sup>2</sup> ✗ •<sup>1</sup> ✓<sub>1</sub> •<sup>3</sup> ✗</p> |                | <p>•<sup>1</sup> ✗ •<sup>2</sup> ✗ •<sup>3</sup> ✗</p> |          |

| Question |     |  | Generic scheme                                                                                                                                                 | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Max mark |
|----------|-----|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 10.      | (a) |  | <p>•<sup>1</sup> use <math>-5</math> in synthetic division or evaluation of quartic</p> <p>•<sup>2</sup> complete division/evaluation and interpret result</p> | <p>•<sup>1</sup></p> $\begin{array}{r rrrrr} -5 & 1 & 3 & -7 & 9 & -30 \\ & & 1 & & & \end{array}$ <p>or</p> $(-5)^4 + 3 \times (-5)^3 - 7 \times (-5)^2 + 9 \times (-5) - 30$ <p>•<sup>2</sup></p> $\begin{array}{r rrrrr} -5 & 1 & 3 & -7 & 9 & -30 \\ & & -5 & 10 & -15 & 30 \\ \hline & 1 & -2 & 3 & -6 & 0 \end{array}$ <p>Remainder = 0 <math>\therefore (x+5)</math> is a factor<br/> <b>OR</b><br/> <math>f(-5) = 0 \therefore (x+5)</math> is a factor</p> | 2        |

#### Notes:

- Communication at •<sup>2</sup> must be consistent with working at that stage i.e. a candidate's working must arrive legitimately at 0 before •<sup>2</sup> can be awarded.
- Accept any of the following for •<sup>2</sup>:
  - ' $f(-5) = 0$  so  $(x+5)$  is a factor'
  - 'since remainder = 0, it is a factor'
  - the '0' from any method linked to the word 'factor' by 'so', 'hence',  $\therefore$ ,  $\rightarrow$ ,  $\Rightarrow$  etc.
- Do not accept any of the following for •<sup>2</sup>:
  - double underlining the '0' or boxing the '0' without comment
  - ' $x = -5$  is a factor', ' $\dots$  is a root'
  - the word 'factor' only, with no link.

#### Commonly Observed Responses:

##### Candidate A - grid method

$$\begin{array}{r|rrrr} x^3 & & & & \\ x & x^4 & -2x^3 & & \\ 5 & 5x^3 & & & \end{array} \quad \bullet^1 \checkmark$$
  

$$\begin{array}{r|rrrr} x^3 & -2x^2 & +3x & -6 \\ x & x^4 & -2x^3 & +3x^2 & -6x \\ 5 & 5x^3 & -10x^2 & +15x & -30 \end{array}$$

with no remainder

$\therefore (x+5)$  is a factor •<sup>2</sup> ✓

##### Candidate B - grid method

$$\begin{array}{r|rrrr} x^3 & & & & \\ x & x^4 & -2x^3 & & \\ 5 & 5x^3 & & & \end{array} \quad \bullet^1 \checkmark$$
  

$$\begin{array}{r|rrrr} x^3 & -2x^2 & +3x & -6 \\ x & x^4 & -2x^3 & +3x^2 & -6x \\ 5 & 5x^3 & -10x^2 & +15x & -30 \end{array}$$

$\therefore (x+5)(x^3 - 2x^2 + 3x - 6) = x^4 + 3x^3 - 7x^2 + 9x - 30$

$\therefore (x+5)$  is a factor •<sup>2</sup> ✓



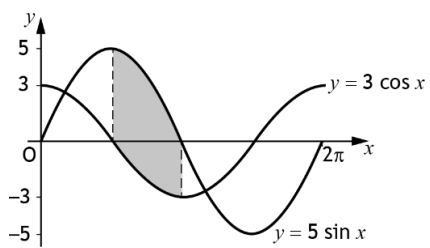
| Question |     | Generic scheme                                                                                                                                                                                                                                  | Illustrative scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Max mark |
|----------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 10.      | (b) | <p>•<sup>3</sup> identify cubic and attempt to factorise</p> <p>•<sup>4</sup> find second factor</p> <p>•<sup>5</sup> identify quadratic</p> <p>•<sup>6</sup> interpret lack of solutions of quadratic</p> <p>•<sup>7</sup> state solutions</p> | <p>•<sup>3</sup> eg</p> $\begin{array}{r rrrr} 1 & 1 & -2 & 3 & -6 \\ & & 1 & -1 & \\ \hline & 1 & -1 & & \end{array}$ <p>or</p> $\begin{array}{r rrrr} 2 & 1 & -2 & 3 & -6 \\ & & 2 & 0 & \\ \hline & 1 & 0 & & \end{array}$ <p>•<sup>4</sup></p> $\begin{array}{r rrrr} 2 & 1 & -2 & 3 & -6 \\ & & 2 & 0 & 6 \\ \hline & 1 & 0 & 3 & 0 \end{array}$ <p>leading to <math>(x-2)</math> or <math>x=2</math></p> <p>•<sup>5</sup> <math>x^2 + 3</math></p> <p>•<sup>6</sup> <math>b^2 - 4ac = -12 &lt; 0</math><br/> <math>\therefore</math> no (further real) solutions<br/> OR<br/> <math>x^2 = -3</math> or <del><math>x^2 = -3</math></del><br/> <math>\therefore</math> no (further real) solutions</p> <p>•<sup>7</sup> <math>x = -5, x = 2</math></p> | 5        |

#### Notes:

- Candidates who arrive at  $(x+5)(x-2)(x^2+3)$  by using algebraic long division or by inspection gain •<sup>3</sup>, •<sup>4</sup> and •<sup>5</sup>.
- Evidence for •<sup>6</sup> may appear in the quadratic formula.
- At •<sup>6</sup> accept interpretations such as “no further roots”, “no solutions” and “cannot factorise further” with justification.
- At •<sup>6</sup> accept  $x = \sqrt{-3}$  leading to “not possible” and “not real”.
- Where there is no reference to  $b^2 - 4ac$  accept ‘ $-12 < 0$  so no real roots’ with the remaining roots stated for •<sup>6</sup> - see candidates E and F.
- Do not accept any of the following for •<sup>6</sup>:
  - $(x+5)(x-2)(x^2+3)$  no further roots/cannot factorise further.
  - $(x+5)(x-2)(\dots)(\dots)(\dots)(\dots)$  no further roots/cannot factorise further.
- Where the quadratic factor obtained at •<sup>5</sup> can be factorised, •<sup>6</sup> and •<sup>7</sup> are not available.
- <sup>7</sup> is only available where •<sup>6</sup> has been awarded.

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Generic scheme                                                             | Illustrative scheme                                                                        | Max mark                                                                         |         |      |      |     |       |         |         |       |   |        |          |        |       |  |       |     |     |     |       |     |     |     |     |     |     |  |       |      |   |     |       |   |      |    |         |   |    |  |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------|------|------|-----|-------|---------|---------|-------|---|--------|----------|--------|-------|--|-------|-----|-----|-----|-------|-----|-----|-----|-----|-----|-----|--|-------|------|---|-----|-------|---|------|----|---------|---|----|--|--|--|
| 10.(continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                            |                                                                                            |                                                                                  |         |      |      |     |       |         |         |       |   |        |          |        |       |  |       |     |     |     |       |     |     |     |     |     |     |  |       |      |   |     |       |   |      |    |         |   |    |  |  |  |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                            |                                                                                            |                                                                                  |         |      |      |     |       |         |         |       |   |        |          |        |       |  |       |     |     |     |       |     |     |     |     |     |     |  |       |      |   |     |       |   |      |    |         |   |    |  |  |  |
| <b>Candidate C</b><br>$(x+5)(x-2)(x^2+3)$<br>$b^2-4ac=0-12<0$<br>so no solutions<br>$x=-5, x=2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $\bullet^5 \checkmark$<br>$\bullet^6 \checkmark$<br>$\bullet^7 \checkmark$ | <b>Candidate D</b><br>$(x+5)(x-2)(x^2+3)$<br>$b^2-4ac<0$<br>so no solutions<br>$x=-5, x=2$ | $\bullet^5 \checkmark$<br>$\bullet^6 \wedge$<br>$\bullet^7 \boxed{\checkmark 2}$ |         |      |      |     |       |         |         |       |   |        |          |        |       |  |       |     |     |     |       |     |     |     |     |     |     |  |       |      |   |     |       |   |      |    |         |   |    |  |  |  |
| <b>Candidate E</b><br>$(x+5)(x-2)(x^2+3)$<br>$-12<0$<br>so no solutions<br>$x=-5, x=2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $\bullet^5 \checkmark$<br>$\bullet^6 \checkmark$<br>$\bullet^7 \checkmark$ | <b>Candidate F</b><br>$(x+5)(x-2)(x^2+3)$<br>$-12<0$<br>so no solutions                    | $\bullet^5 \checkmark$<br>$\bullet^6 \wedge \bullet^7 \wedge$                    |         |      |      |     |       |         |         |       |   |        |          |        |       |  |       |     |     |     |       |     |     |     |     |     |     |  |       |      |   |     |       |   |      |    |         |   |    |  |  |  |
| <b>Candidate G - grid method</b><br>(a)<br><table><tr><td></td><td><math>x^3</math></td><td><math>-2x^2</math></td><td><math>3x</math></td><td><math>-6</math></td></tr><tr><td><math>x</math></td><td><math>x^4</math></td><td><math>-2x^3</math></td><td><math>+3x^2</math></td><td><math>-6x</math></td></tr><tr><td>5</td><td><math>5x^3</math></td><td><math>-10x^2</math></td><td><math>+15x</math></td><td><math>-30</math></td></tr></table><br>(b)<br><table><tr><td></td><td><math>x^2</math></td><td>...</td><td>...</td></tr><tr><td><math>x</math></td><td><math>x^3</math></td><td>...</td><td>...</td></tr><tr><td>...</td><td>...</td><td>...</td><td>...</td></tr></table><br>$\bullet^3 \checkmark$<br><p><math>\bullet^3</math> is awarded for evidence of the cubic expression (which may be in the grid from part (a) ) <b>AND</b> the terms in the diagonal boxes summing to the second and third terms in the cubic respectively.</p> <table><tr><td></td><td><math>x^2</math></td><td><math>0x</math></td><td>3</td></tr><tr><td><math>x</math></td><td><math>x^3</math></td><td>0</td><td><math>3x</math></td></tr><tr><td>-2</td><td><math>-2x^2</math></td><td>0</td><td>-6</td></tr></table><br>$\bullet^4 \checkmark$<br>$(x+5)(x-2)(x^2+3)$<br>$b^2-4ac=-12<0$<br>$\therefore$ no more solutions<br>$x=-5, x=2$<br>$\bullet^5 \checkmark$<br>$\bullet^6 \checkmark$<br>$\bullet^7 \checkmark$ |                                                                            |                                                                                            | $x^3$                                                                            | $-2x^2$ | $3x$ | $-6$ | $x$ | $x^4$ | $-2x^3$ | $+3x^2$ | $-6x$ | 5 | $5x^3$ | $-10x^2$ | $+15x$ | $-30$ |  | $x^2$ | ... | ... | $x$ | $x^3$ | ... | ... | ... | ... | ... | ... |  | $x^2$ | $0x$ | 3 | $x$ | $x^3$ | 0 | $3x$ | -2 | $-2x^2$ | 0 | -6 |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $x^3$                                                                      | $-2x^2$                                                                                    | $3x$                                                                             | $-6$    |      |      |     |       |         |         |       |   |        |          |        |       |  |       |     |     |     |       |     |     |     |     |     |     |  |       |      |   |     |       |   |      |    |         |   |    |  |  |  |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $x^4$                                                                      | $-2x^3$                                                                                    | $+3x^2$                                                                          | $-6x$   |      |      |     |       |         |         |       |   |        |          |        |       |  |       |     |     |     |       |     |     |     |     |     |     |  |       |      |   |     |       |   |      |    |         |   |    |  |  |  |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $5x^3$                                                                     | $-10x^2$                                                                                   | $+15x$                                                                           | $-30$   |      |      |     |       |         |         |       |   |        |          |        |       |  |       |     |     |     |       |     |     |     |     |     |     |  |       |      |   |     |       |   |      |    |         |   |    |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $x^2$                                                                      | ...                                                                                        | ...                                                                              |         |      |      |     |       |         |         |       |   |        |          |        |       |  |       |     |     |     |       |     |     |     |     |     |     |  |       |      |   |     |       |   |      |    |         |   |    |  |  |  |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $x^3$                                                                      | ...                                                                                        | ...                                                                              |         |      |      |     |       |         |         |       |   |        |          |        |       |  |       |     |     |     |       |     |     |     |     |     |     |  |       |      |   |     |       |   |      |    |         |   |    |  |  |  |
| ...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | ...                                                                        | ...                                                                                        | ...                                                                              |         |      |      |     |       |         |         |       |   |        |          |        |       |  |       |     |     |     |       |     |     |     |     |     |     |  |       |      |   |     |       |   |      |    |         |   |    |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $x^2$                                                                      | $0x$                                                                                       | 3                                                                                |         |      |      |     |       |         |         |       |   |        |          |        |       |  |       |     |     |     |       |     |     |     |     |     |     |  |       |      |   |     |       |   |      |    |         |   |    |  |  |  |
| $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $x^3$                                                                      | 0                                                                                          | $3x$                                                                             |         |      |      |     |       |         |         |       |   |        |          |        |       |  |       |     |     |     |       |     |     |     |     |     |     |  |       |      |   |     |       |   |      |    |         |   |    |  |  |  |
| -2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $-2x^2$                                                                    | 0                                                                                          | -6                                                                               |         |      |      |     |       |         |         |       |   |        |          |        |       |  |       |     |     |     |       |     |     |     |     |     |     |  |       |      |   |     |       |   |      |    |         |   |    |  |  |  |



| Question                     |     |  | Generic scheme                                                                                     | Illustrative scheme                                                                                                                                 | Max mark |
|------------------------------|-----|--|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 11.                          | (b) |  | <ul style="list-style-type: none"> <li>•<sup>4</sup> identify boundaries and shade area</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>4</sup>  </li> </ul> | 1        |
| Notes:                       |     |  |                                                                                                    |                                                                                                                                                     |          |
|                              |     |  |                                                                                                    |                                                                                                                                                     |          |
| Commonly Observed Responses: |     |  |                                                                                                    |                                                                                                                                                     |          |
|                              |     |  |                                                                                                    |                                                                                                                                                     |          |

| Question |  |  | Generic scheme                                                                                                                                                                    | Illustrative scheme                                                                                                                                                                            | Max mark |
|----------|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 12.      |  |  | <b>Method 1</b><br>• <sup>1</sup> identify common factor<br><br>• <sup>2</sup> complete the square<br><br>• <sup>3</sup> process for $c$ and write in required form               | <b>Method 1</b><br>• <sup>1</sup> $-2(x^2 + 6x \dots$ stated or implied by • <sup>2</sup><br>• <sup>2</sup> $-2(x+3)^2 \dots$<br>• <sup>3</sup> $-2(x+3)^2 + 25$                               | 3        |
|          |  |  | <b>Method 2</b><br>• <sup>1</sup> expand completed square form<br><br>• <sup>2</sup> equate coefficients<br><br>• <sup>3</sup> process for $b$ and $c$ and write in required form | <b>Method 2</b><br>• <sup>1</sup> $ax^2 + 2abx + ab^2 + c$ stated or implied by • <sup>2</sup><br>• <sup>2</sup> $a = -2, 2ab = -12,$<br>and $ab^2 + c = 7$<br>• <sup>3</sup> $-2(x+3)^2 + 25$ |          |

#### Notes:

- $-2(x+3)^2 + 25$  with no working gains •<sup>1</sup> and •<sup>2</sup> only. However, see Candidate E.
- <sup>1</sup> and •<sup>3</sup> are not available in cases where  $a > 0$ . For example, see Candidate F.

#### Commonly Observed Responses:

|                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Candidate A</b><br>$-2(x^2 + 6) + 7$<br>$-2((x+3)^2 - 9) + 7$ • <sup>1</sup> ✓    • <sup>2</sup> ✓<br>$-2(x+3)^2 + 25$ • <sup>3</sup> ✓<br>See the exception to marking principle (h)                                                 | <b>Candidate B</b><br>$ax^2 + 2abx + ab^2 + c$ • <sup>1</sup> ✓<br>$a = -2, 2ab = -12, ab^2 + c = 7$ • <sup>2</sup> ✓<br>$b = 3, c = 25$ • <sup>3</sup> ^<br><div>             •<sup>3</sup> is lost as answer is not in completed square form           </div> |
| <b>Candidate C</b><br>$-2(x^2 + 12x) + 7$ • <sup>1</sup> ✗<br>$-2((x+6)^2 - 36) + 7$ • <sup>2</sup> <input checked="" type="checkbox"/> <sub>1</sub><br>$-2(x+6)^2 + 79$ • <sup>3</sup> <input checked="" type="checkbox"/> <sub>1</sub> | <b>Candidate D</b><br>$-2((x+6)^2 - 36) + 7$ • <sup>1</sup> ✗    • <sup>2</sup> ✗<br>$-2(x+6)^2 + 79$ • <sup>3</sup> <input checked="" type="checkbox"/> <sub>1</sub>                                                                                           |
| <b>Candidate E</b><br>$-2(x+3)^2 + 25$ • <sup>1</sup> ✓    • <sup>2</sup> ✓<br>Check: $= -2(x^2 + 6x + 9) + 25$<br>$= -2x^2 - 12x - 18 + 25$<br>$= -2x^2 - 12x + 7$ • <sup>3</sup> ✓                                                     | <b>Candidate F</b><br>$-2x^2 - 12x + 7$<br>$= 2x^2 + 12x - 7$ • <sup>1</sup> ✗<br>$= 2(x^2 + 6x \dots$<br>$= 2(x+3)^2 \dots$ • <sup>2</sup> <input checked="" type="checkbox"/> <sub>1</sub><br>$= -2(x+3)^2 \dots$ • <sup>3</sup> ✗                            |

| Question |     |      | Generic scheme                                                                     | Illustrative scheme                                                  | Max mark |
|----------|-----|------|------------------------------------------------------------------------------------|----------------------------------------------------------------------|----------|
| 13.      | (a) | (i)  | • <sup>1</sup> state exact value                                                   | • <sup>1</sup> $\sqrt{3}$                                            | 1        |
|          |     | (ii) | • <sup>2</sup> interpret notation<br>• <sup>3</sup> state expression for $f(g(x))$ | • <sup>2</sup> $f(2x)$ or $2\sin(g(x))$<br>• <sup>3</sup> $2\sin 2x$ | 2        |

**Notes:**

- For  $f(g(x)) = 2\sin 2x$  without working, award both •<sup>2</sup> and •<sup>3</sup>.
- Working for (a)(ii) may be found in (a)(i)

**Commonly Observed Responses:**

|                                                                                                                         |                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Candidate A</b><br>(a)(ii) $f(g(x)) = 4\sin x$ • <sup>2</sup> ✗   • <sup>3</sup> <input checked="" type="checkbox"/> | <b>Candidate B - Beware of “2 attempts”</b><br>$f(g(x)) = 2\sin x$ • <sup>2</sup> ✗   • <sup>3</sup> ✗<br>$f(2x) = 2\sin 2x$ |
|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|

|  |     |      |                                                                                                                                                                    |                                                                                                                                                                                                                                                        |   |
|--|-----|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|  | (b) | (i)  | • <sup>4</sup> find the value of $\sin p$                                                                                                                          | • <sup>4</sup> $\frac{1}{6}$                                                                                                                                                                                                                           | 1 |
|  |     | (ii) | • <sup>5</sup> expand $f(g(p))$ using double angle formula<br><br>• <sup>6</sup> find value of $\cos p$<br><br>• <sup>7</sup> substitute and determine exact value | • <sup>5</sup> $2 \times 2\sin p \cos p$ or $4\sin p \cos p$ <b>stated explicitly</b><br><br>• <sup>6</sup> $\frac{\sqrt{35}}{6}$<br><br>• <sup>7</sup> $2 \times 2 \times \frac{1}{6} \times \frac{\sqrt{35}}{6}$<br>leading to $\frac{\sqrt{35}}{9}$ | 3 |

**Notes:**

- <sup>5</sup> is not available for expansions which do not involve  $p$ . •<sup>6</sup> and •<sup>7</sup> are still available.  
However, accept  $\sin^{-1}\left(\frac{1}{6}\right)$  in place of  $p$  - see Candidate C.
- <sup>7</sup> is only available as a consequence of substituting into a valid formula from •<sup>5</sup>.
- Do not penalise trigonometric ratios which are less than -1 or greater than 1 throughout this question.

**Commonly Observed Responses:**

|                                                                                                                                                                                                                                                                          |  |  |  |  |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
| <b>Candidate C</b><br>$f(g(p)) = 4\sin\left(\sin^{-1}\left(\frac{1}{6}\right)\right)\cos\left(\sin^{-1}\left(\frac{1}{6}\right)\right)$ • <sup>5</sup> ✓<br>$4 \times \frac{1}{6} \times \frac{\sqrt{35}}{6}$ • <sup>6</sup> ✓<br>$\frac{\sqrt{35}}{9}$ • <sup>7</sup> ✓ |  |  |  |  |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|

[END OF MARKING INSTRUCTIONS]